

Multivariate Adaptive Autoregressive Modeling and Kalman Filtering for Motor Imagery BCI

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Abstract—Adaptive autoregressive (AAR) modeling of the EEG time series and the AAR parameters has been widely used in Brain computer interface (BCI) systems as input features for the classification stage. Multivariate adaptive autoregressive modeling (MVAAR) also has been used in literature. This paper revisits the use of MVAAR models and propose the use of adaptive Kalman filter (AKF) for estimating the MVAAR parameters as features in a motor imagery BCI application. The AKF approach is compared to the alternative short time moving window (STMW) MVAAR parameter estimation approach. Though the two MVAAR methods show a nearly equal classification accuracy, the AKF possess the advantage of higher estimation update rates making it easily adoptable for on-line BCI systems.

Keywords—Brain Computer Interface, Motor Imagery, Adaptive Kalman filter, Multivariate Autoregressive Modeling

I. INTRODUCTION

Brain Computer Interface (BCI) systems provide a communication channel between the human and the computer, through the brain waves measured using the brain imaging modality electroencephalography (EEG). Human thoughts measured through EEG is passed through various signal processing stages to convert to a control signal. The main application of the BCI systems is to restore sensory and motor functions in patient with neural disorders and motor disabilities. A BCI system essentially involves signal processing steps such as data preprocessing, feature extraction and feature classification. Commonly used EEG features include changes in power of certain frequency bands, fast fourier transform (FFT), autoregressive (AR) parameters and wavelet coefficients [1]. In this paper we focus on AR parameters as features for EEG-based motor imagery BCI application.

When using AR models to fit data, the stationarity is necessary for accurate estimation of AR model parameters. However, in a typical cognitive experiment the brain undergoes rapid transitions in its functional states, thus creating non-stationarity in the recorded EEG data. Thus, adaptive estimation techniques should be used when estimating the AR model parameters. Adaptive AR (AAR) modeling is the

alternate to AR modeling of the EEG signals. AAR parameters can capture the time-varying second order statistical moments from the EEG signals. AAR has been proposed and used in EEG-based BCI's in many instances [2]–[5]. Improved multivariate adaptive autoregressive (MVAAR) models have also been proposed in [6]–[8] to extract features from multi-channel EEG signals in BCI applications.

The AAR modeling performs feature extraction channel by channel and concatenate the features to a single feature vector. This procedure will only consider the relationship between the present signal and the past signal of every channel itself and neglects the cross-talking between the EEG channel, which may carry important features about the performed task. In contrast the MVAAR models will capture the interaction between the channel in the MVAAR parameters and hence proved as better compared to the AR and AAR parameters when used as features in motor imagery BCI applications [6]–[8]. In [7] the authors use AAR and MVAAR approach using the segmentation process to extract features from a motor imagery dataset. They perform a bi-class classification between the left and right motor imagery using the Mahalanobis distance for the discrimination. They reported a maximum accuracy of 88.57% when using MVAAR compared to the maximum accuracy of 84.29% when AAR is used, showing that MVAAR is better than AAR.

Segmentation and adaptive filtering are the two main adaptive methods used to handle the non-stationarities in the EEG recordings. In segmentation the EEG signal is divided in to short windows that are locally stationary and the estimation algorithms is applied to moving windows [2]. We will refer this as the short-time moving window (STMW) approach hereafter. MVAAR parameters are estimated using robust multivariate least squares and non-least squares algorithms such as the multivariate Burg-type method [9], Levinson-Wiggins-Robinson (LWR) or the multichannel Levinson-Durbin recursion, which is the multivariate extension the univariate Levinson recursion [10] and step-wise least squares in ARFIT [11]. The state-space formulation of the MVAAR model enables the use of the

adaptive filters in the parametric analysis of nonstationary EEG data. Many different adaptive filtering algorithms are available, such as least means squares (LMS), recursive least squares (RLS) and Kalman filtering [12].

In the standard Kalman filter recursions, the knowledge of state noise (Σ_V) and process noise (Σ_E) is critical for precise estimate calculations. Due to the non-stationarity of the EEG signals Σ_E and Σ_V will be time varying parameters and with no *a priori* knowledge available should be recursively estimated within the standard Kalman filter iterations. When the noise covariance estimations are embedded in the standard Kalman filter recursions it is referred to as an adaptive Kalman filter (AKF) Algorithm.

The advantage of using the AKF is that the time resolution of the computed parameters is equal to the sampling rate of the system as the Kalman filter updates on a sample by sample basis. Thus, the real time requirements in an on-line BCI system is easily fulfilled. On the other hand, if a STMW approach was to be used as the adaptive technique, this will require much higher computational effort to obtain the same update rate [2]. Further, the Kalman filter is regarded as the optimal estimator of time varying parameters of an AAR model in a mean square sense. Therefore, the work presented in this paper will consider the AKF as the method of choice for extracting the MVAAR features from the EEG signals.

The motor imagery data set used in our work [13], has been analysed with AAR and MVAAR parameters (as features) in earlier work [5], [7], [14]. In [5], [14] the authors use AKF to extract AAR parameters to classify of motor imagery data while the work in [7] uses STMW approach to extract MVAAR features. Studies reported in [6]–[8] report the use of MVAAR modeling of the EEG signals to extract features for the BCI systems. Aforementioned studies mainly present a comparison between using the AAR and MVAAR features on the classification accuracy. On the other hand we present a comparison on using the STMW and the AKF for feature extraction using MVAAR modeling. Furthermore, we present a complete description on utilising the AKF with the MVAAR for formulation of the feature vector in contrast to the work presented in [8].

Once the features are extracted from EEG signals, the next step is to determine to which class the features belong to. This is referred to as feature classification. A number of widely used classifiers are Linear discriminant analysis (LDA), K-nearest-neighbor (kNN) algorithms, Support vector machine (SVM), and Neural networks (NN) [15]. In our analysis we use three widely used classifiers, namely kNN, LDA and SVM.

The rest of the paper is organised as follows. Section 2 describes the experimental data set used in our paper. The MVAAR parameter-based feature vector formulation is presented step by step in section 3. Section 4 presents the results and a discussion of the findings of the study and section 5 concludes the paper.

II. THE EXPERIMENTAL DATA SET

The motor imagery data analysed in this paper are from the publicly available BCI competition 2003 data set III, provided by the Department of Medical Informatics, Institute for Biomedical Engineering, University of Technology, Graz, Austria [13]. Here we provide a brief description of the data set. The detailed description can be found in [13].

The EEG signals were recorded from a 25 year old healthy female. The subject was asked to control a feedback bar by imagination of left and right hand movement, while seated relaxing on a chair with armrest. EEG recording were taken from 3 bipolar channels from the positions C3, C4 and Cz. Each trial lasted 9s. After 2s's of silence, an acoustic stimulation was presented at $t = 2s$ indicating the beginning of the trial. The visual feedback with a cross '+' was presented from 2s to 3s. At $t = 3s$ an arrow pointing left or right was displayed as a cue. The subject was asked to move the feedback bar to the direction of the cue by imagination of left or right hand movement. The AAR parameters calculated from channels C3 and C4 were combined with a discriminant analysis into one output parameter to provide the feedback.

The experiment consists of 7 runs each consisting of 40 trials. This resulted in a data set with 280 trials of 9s length. The data was recorded on the same day with several minutes of break in between the runs. The sampling rate was 128 Hz and the was filtered between 0.5 and 30 Hz. The trials for training and testing were randomly selected. The training data set combined 140 trials. The class labels, '1' and '2', corresponded to left and right, respectively. The test data set contained 140 trials, each belonging to either class '1' or class '2'.

III. FEATURE EXTRACTION

A. Multivariate Adaptive Autoregressive Modeling of the EEG time series

The MVAAR model with model order p for k signal sources can be described by the following equation:

$$Y(n) = \sum_{i=1}^p A_i(n)Y(n-i) + E(n) \quad (1)$$

where $Y(n) = [y_1(n), y_2(n) \dots y_k(n)]^T$, y_j denotes the j^{th} signal source and $n = 1, 2, \dots, L$ denotes a time point. The superscript T refers to the transpose of a matrix throughout the paper. L is the total number of time points in the data. $A_i(n)$ is the $(k \times k)$ model coefficient matrix at lag $i = 0, \dots, p$. $E(n) = [e_1(n), e_2(n), \dots, e_k(n)]^T$ is a vector of zero mean white noise input with covariance matrix Σ_E .

1) *State space formulation of the MVAAR model:* In order to apply a Kalman filtering approach to estimate the MVAR coefficients, the model (1) should be represented in a state space formulation [16],

$$Y(n) = C(n)a(n) + E(n) \quad (2)$$

where,

$a(n) = [\text{vec}[A_1^T(n)]^T, \text{vec}[A_2^T(n)]^T, \dots, \text{vec}[A_p^T(n)]^T]^T$ is the concatenation of vectorised $\{A_i(n)\}_{i=1}^p$ of size $(k^2p \times 1)$,

$$A_i(n) = \begin{pmatrix} a_{11}^i(n) & a_{12}^i(n) & \dots & a_{1k}^i(n) \\ a_{21}^i(n) & a_{22}^i(n) & \dots & a_{2k}^i(n) \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1}^i(n) & a_{k2}^i(n) & \dots & a_{kk}^i(n) \end{pmatrix},$$

and

$$C(n) = [I_k \otimes Y^T(n-1), \dots, I_k \otimes Y^T(n-p)]$$

is the concatenated matrix of the past measurements of size $k \times k^2P$ with $\otimes$ is the Kronecker-product and I_k is the $k \times k$ identity matrix. In the state space formulation in equation (2), the vector $a(n)$ contains the states that has to be estimated. In order to have a complete model to be applied with the Kalman filter, a state model is required. When *a priori* knowledge is not available about the states the most common approach is to use a random walk model [16]. Together the random walk state variation and equation (2) will constitute a linear state space model,

$$\begin{aligned} a(n) &= a(n-1) + V(n) \\ Y(n) &= C(n)a(n) + E(n) \end{aligned} \quad (3)$$

where $V(n)$ is the state noise, which is a multivariate Gaussian noise with zero mean and covariance matrix Σ_V .

B. Adaptive Kalman filter for MVAAR parameter estimation

The MVAAR parameters can be recursively estimated in response to dynamic changes in the signals. At each time instant n , a new estimate of $a(n)$, $\hat{a}(n)$ may be obtained using the Kalman filter recursions. In order to estimate the noise covariances Σ_E and Σ_V recursively within the standard Kalman filter iterations adaptive estimation techniques should be used. Among the many different adaptive estimation techniques [12], we use two update equation from previous work [17],

$$\begin{aligned} \Sigma_E(n) &= \alpha \Sigma_E(n-1) + (1-\alpha) \tilde{Y} \tilde{Y}^T \\ \Sigma_V(n) &= I(1-\alpha) \text{trace}(P(n|n))/P \end{aligned} \quad (4)$$

where α is the forgetting factor that controls the adaptation speed with its values lying in the interval $[0, 1]$ and is chosen to be near to 1 [18], $\tilde{Y}(n)$ measurement residual and $P(n|n)$ is the *a posteriori* estimation error covariance matrix.

The multivariate AKF recursions for the system (3) is given below in two phases;

- Time Update

$$\begin{aligned} \hat{a}(n|n-1) &= \tilde{a}(n-1|n-1) \\ P(n|n-1) &= P(n-1|n-1) + \Sigma_V(n) \\ \tilde{Y}(n) &= Y(n) - C(n)\hat{a}(n|n-1) \end{aligned} \quad (5)$$

- Measurement Update

$$\begin{aligned} \Sigma_E(n) &= \alpha \Sigma_E(n-1) + (1-\alpha) \tilde{Y} \tilde{Y}^T \\ S(n) &= C(n)P(n|n-1)C(n)^T \\ &\quad + \Sigma_E(n) \\ K(n) &= P(n|n-1)C(n)^T S(n)^{-1} \\ \hat{a}(n|n) &= \hat{a}(n|n-1) + K(n)\tilde{Y}(n) \\ P(n|n) &= [I - K(n)C(n)]P(n|n-1) \\ \Sigma_V(n) &= I(1-\alpha) \text{trace}(P(n|n))/p \end{aligned} \quad (6)$$

where $\hat{a}(n|n-1)$, $P(n|n-1)$ are the *a priori* state estimate and the estimation error covariance matrix respectively, $K(n)$ is the Kalman filter gain and $S(n)$ the residual covariance matrix.

C. Formulation of the feature vectors for the experimental data

The EEG features used in this study are based on the MVAAR parameters estimated by STMW and the AKF. The channels C3 and C4 are only used in the analysis. Each trial in the training data is bandpass filtered at 9-30Hz. The MVAAR model is parameterised with the model order and from analysis done off-line with the training data the optimal model order was selected as $p = 13$ using the Schwarz Bayesian Criterion (SBC). This resulted in 52 features at each iteration. In the STMW approach the window length was selected as 1 second which resulted in 128 samples. The window was moved by one sample to estimate the MVAAR parameters. This resulted in 1025 windows each with 52 features. In the AKF approach the MVAAR parameters were estimated for each time point in a trial of 9s at 128Hz, resulting in 1152 feature vectors.

The MVAR parameters were calculated using the step-wise least squares algorithm [19] implemented in ARFIT [11]. For the AKF the initial values used were $a(0) = 0$, $P(0|0) = 0$, $\sigma_E = I_k$ and $\sigma_V = (1-\alpha)I_k$.

The feature vector F for a given time point n is formulated as $F(n) = [a_{11}^1(n), \dots, a_{1k}^1(n), \dots, a_{k1}^1(n), \dots, a_{kk}^1(n), \dots, a_{11}^p(n), \dots, a_{1k}^p(n), \dots, a_{k1}^p(n), \dots, a_{kk}^p(n)]$.

IV. RESULTS AND DISCUSSION

The feature vector $F(n)$ is extracted for moving window (STMW) or a time point (AKF) of each trial of the training and testing data. The kNN, LDA and SVM classifiers were used to evaluate the classification accuracy for each evaluation point. Fig. 1 shows the classification accuracy for the STMW and Fig. 2 shows the classification accuracy with different classifiers for AKF feature extractions.

The maximum accuracy of 84.29% classification occurs in the interval $t = 4 - 5$ s in the STMW approach while in the AKF approach the maximum accuracy of 85.71% occurs in the interval $t = 6 - 7$ s. The maximum accuracy is attained with the SVM classifier and the LDA classifier

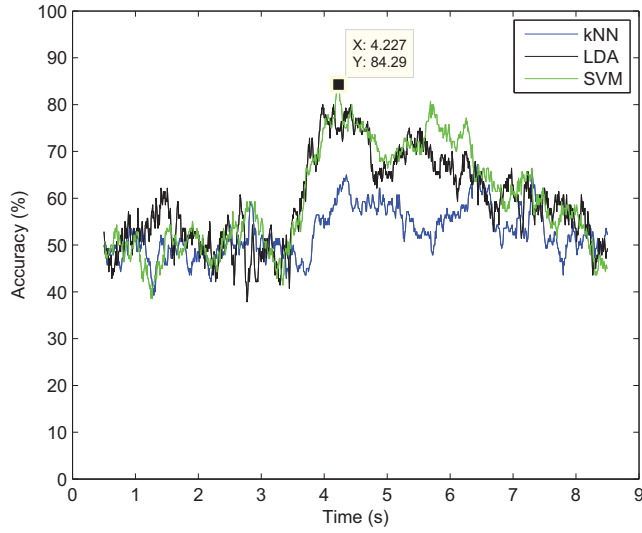


Figure 1. Classification accuracy for each moving window for STMW-based feature extraction

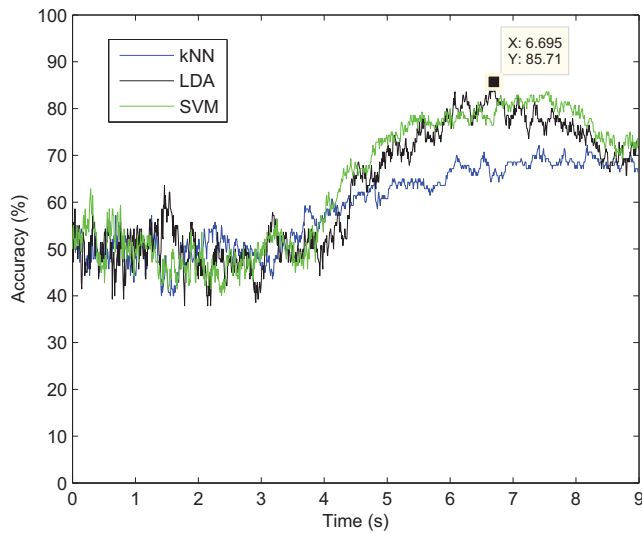


Figure 2. Classification accuracy for each moving window for AKF-based feature extraction

in STMW and AKF respectively. This maximum accuracy intervals can also be validated with the findings related to event-related desynchronisation (ERD) and event-related synchronisation (ERS) on the same data set [7] and our analysis shown in Fig.3. The Fig.3 shows that the α and β band (9-30Hz) amplitude separation varies significantly during the time interval $t = 4 - 9$ s between the two channels C3 and C4 reaching a maximum in the time interval $t = 4 - 5$ s. This means that the MVAAR based features

extracted from the signals in the frequency band 9-30Hz has been able to capture this separability between the two classes left and right motor imagery which is validated by the ERD/ERS analysis shown in Fig.3. It can be seen that the MVAAR model precisely captures the spectral components of the EEG.

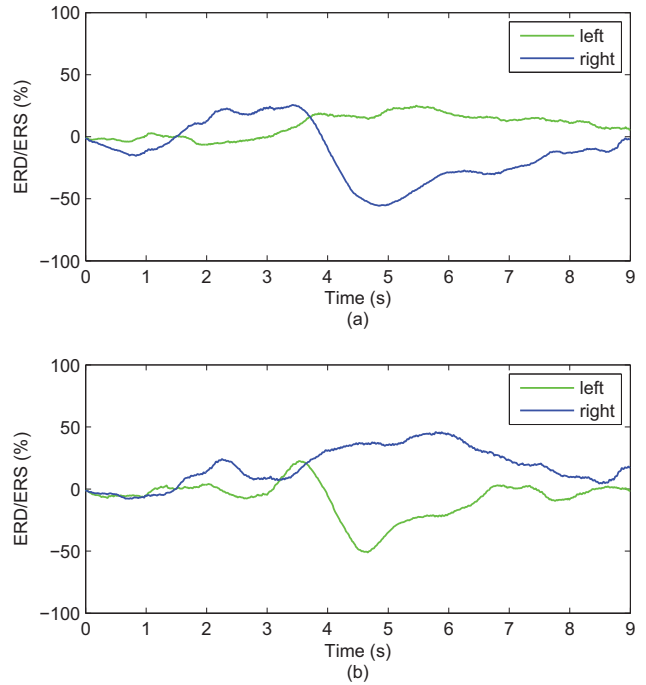


Figure 3. ERD/ERS of the motor imagery data (a) Channel C3 (b) Channel C4 in the 9-30Hz α and β bands. Contralateral ERD and ipsilateral ERS is observed in the channels. The separation among the left and right movement imaginary tasks occurs in the time interval $t = 4 - 9$ for both the channels.

V. CONCLUSION

The work presented in this paper discussed the use of MVAAR model parameters extracted with the STMW and AKF approaches. The AKF-based feature extraction is emphasised and the mathematical formulation of the algorithm is presented in the paper with the aim of adopting for on-line BCI systems. Compared to the STMW approach the AKF shows only a marginal increase in the classification accuracy of the motor imagery data. However, has promising extension to be adapted for on-line BCI systems due its inherent sample by sample high update rate. Extensions of the presented work will evaluate the robustness of the MVAAR and AKF-based features for multi-class BCI applications.

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